

Ant Colony Optimization for Aircraft Landings

Ihsen FARAH, Ali KANSOU

University of Le Havre

Laboratory of Applied Mathematics
of Le Havre

Email: ihsen.farah@univ-lehavre.fr

ali.kansou@yahoo.fr

Adnan YASSINE

University of Le Havre

Superior Institute for Logistics Studies
Laboratory of Applied Mathematics

of Le Havre

Email: adnan.yassine@univ-lehavre.fr

Thierry GALINHO

University of Le Havre

Superior Institute for Logistics Studies
Computer Science, Information Processing,
and Systems Laboratory

Email: thierry.galinho@univ-lehavre.fr

Abstract—We address the problem of determining the arrival sequencing and scheduling for aircrafts in air traffic control system. The overall objective of this problem is the minimization of delay cost. The problem is NP-hard for this reason solving the problem optimally is limited to instances of small sizes. In this paper we treat the static case of ASS problem where all data are known in advance. We begin by proposed a mathematical quadratic integer model, with the constraint of limited time window. After we propose a change of variable in order to make linear the initial quadratic constraints. The second target of this work is to resolve the model by an ant colony algorithm. Numerical simulations are applied to real data to show the effectiveness of our new formulation and our approach. They are presented at the end of this article.

Keywords :Air Traffic Control (ATC), Air-Ground Control (AGC), Quadratic Integer Program (QIP), Lienar Integer Program (LIP), Ant Colony Optimization (ACO).

I. INTRODUCTION

To ensure safe, coherent and effective traffic flow, the air controllers provide a service for control traffic. The Air Traffic Control (ATC) is a service provided by controllers, based on ground, who coordinate the movement of aircraft on the ground and in the sky. The ATC services are provided in much of the airspace and are available to all users. The ATC is composed by three categories:

- Control on the ground: represent the control of the aircraft on part of the airport platforms, this subject is well presented in [6].
- The air-ground control deals with the zones surrounding the airport platforms. The aim of this control is to organize the descent of the aircraft for landing, and also their lift, after takeoff. Ground-air control or tower control is responsible for the active runway surfaces. Tower control clears aircraft for takeoff or landing, ensuring that prescribed runway separation exists at all times. If tower control detects any unsafe conditions, a landing aircraft may be told to go-around and be re-sequenced into the landing pattern by the approach or terminal area controller. In Figure 1 the part circled in red represent the approach zone and will be the subject

of this study.

- Air control (En route control): represents the least visible face of air control. However, it is also the phase of control by far the longest for a commercial flight, and which requires a greater number of controllers. It consists in conveying the aircraft from its exit to the approach zone of departure until reaching the exit towards the approach zone of destination. During the length of this period the aircraft will be rooted most correspondingly compared to its flight plan, in certain cases, and because safety, the controllers change the trajectories of the flights. (see [4] for more information about en route control)

In this work, we are interested in the air-ground control to resolve the aircraft landings Problem. This problem consists at obtaining the sequencing order of aircrafts and the scheduling of landing times. The objective to solve the problem of aircraft landings is to minimize the total delay costs.

The landing phase is a very critical phase of the flight. Upon entry to the horizon radar of tower control of an airport, a flight request its permission to land. A date earlier and another at the latest (which depend, among other, on the maximum and minimum speed of the aircraft) will be assigned to the flight. These dates, in any case, are not exceeded; otherwise, the aircraft may be told to go-around and will be reinserted into the landing sequence later. Inside this interval, there is the target landing time which corresponds to the landing time preferred by a flight since this date corresponds to arrival time for which the flight with its cruising speed which is the most economic speed.

Several factors such as the availability of a landing strip, are to be taken into account at the landing time of a flight, for that most of the time, we find a shift (an advance or a delay) between the target landing time and the scheduled arrival time. A delay or an advance compared to the scheduled arrival time causes disturbances at the airport. For better managing these disturbances, a penalty is associated with each delay or advances of a flight. The goal of this work is thus to minimize the total delay time of all flights. in our work, the most important constraints in the reality are taken into

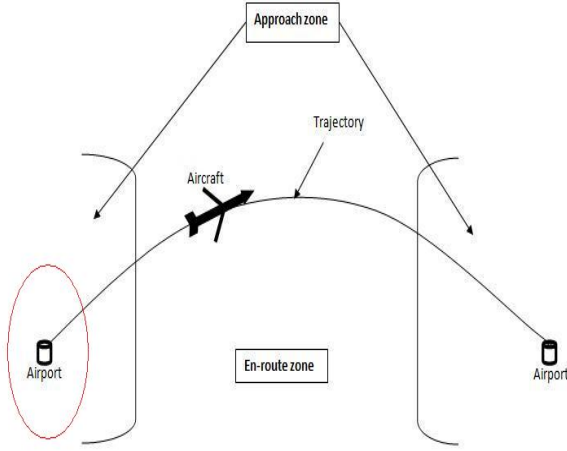


Fig. 1. Air space

account: all flights are landing, each flight must respect its time window of arrival and departure, an interval of separation is imposed between two successive landings (the time of separation varies according to weight, speed, capacity and other technical characteristics of flight), and finally, since in general the Effective Arrival Time (EAT) is greater than the Scheduled Arrival Time (SAT), we will require that.

The main target of this study is to provide a modelling for the flexibility (respecting the time window of aircraft landing time, constraints (9)) and predictability (minimizing the delay to land) of the ASS problem in Air Traffic Control (ATC) system as there are two main indicators for the performance of a system. For the safety of the flights, a time of separation between two successive landings is imposed which depends on several factors such as the types of two planes. The ASS problem has attracted many researchers. The problem in static case of single-runway has been studied by J. E. Beasley et al. [9], who presented a formulation of the problem as a mixed linear program. They used a method based on relaxation of binary variables by introducing additional constraints. Two methods to resolve this problem were presented by A. T. Ernst et al. in [1], an exact method based on the principle of Branch and Bound and an heuristic based on the genetic algorithm. J. Abela and al [5] also presented a branch and bound algorithm for the resolution for this problem. G. B. Cheikh et al. [2] has treated the ASS problem, they have modelled the problem as a Job Shop with partial order and alternative sequences and they proposed an hybrid method combining Genetic Algorithms and Ant Colony Optimization for the resolution. H. Pinol et al [3] has applied the scatter search and Bionomics algorithm for this problem with multi-runway. They considered the linear and nonlinear objective function, and they scored for instances of up to 500 aircraft and five runways. Bianco et al. [10] was also adopted an approach based on a dynamic programming formulation and two heuristics. Numerical results are presented for a number

of randomly generated problems. A particular case is treated in [7], where the authors used one Heuristics based on a population say population heuristics to manage the landing at the airport Heathrow in London. They base on the same Algorithm to resolve the ASS problem in the dynamic case [8].

The most accurate methods found in the literature employ the Branch and Bound method for solving the problem. However the computation time increases exponentially with the size of problems. Therefore researchers have turned to meta-heuristics to solve big sizes instances.

This paper is organized as follow. In section 2, we present the quadratic formulation proposed and we present a change of variables to linearize the quadratic constraints, the prove of this theorem are presented. In section 3, we present our ant colony algorithm to solve the aircrafts landing problem. Numerical simulations are applied to real data involving up to 150 planes are presented in section 4. Finally, we conclude and we give some perspectives for this work in section 5.

II. MODELLING AND RESOLUTION OF ASS PROBLEM

This section is divided in two parts. Firstly, we will construct the directly quadratic model and we will transform the model into linear integer formulation using change of variables. Secondly, we propose an exact method based on branch and bound algorithm and we will present a numerical simulation at the end of this part.

A. Modeling of ASS problem

1) The proposed quadratic model for the ASS problem :

We use two sets of decision variables. The first is a binary variable defined as follow:

$$x_{ij} = \begin{cases} 1, & \text{if the aircraft } j \text{ is landed after } i \\ 0, & \text{otherwise} \end{cases}$$

We note that $x_{ii} = 1$ means i is the first aircraft to land. The second variable used is presented by EAT_i which is the Effective Arrival Time affected to aircraft i , where EAT_i is integer $\forall i = 1, \dots, N_A$ and N_A is the total number of aircraft. note also that S_{ij} is the permission interval of time for aircraft j to follow an aircraft i to land. We propose that a time window of each aircraft i is $[SAT_i; SAT_{imax}]$ where SAT_i and SAT_{imax} are known in advance and respectively they are the Scheduled Arrival Time and the maximum permission scheduled arrival time to land.

The objective of the ASS problem in operating day is to find an optimal arrival sequence respecting the constraint (2) to (10) such that the total airborne delay can be minimized. Then the completely quadratic model based on the two variables x_{ij} and EAT_i is given as follow:

$$\text{Min } \sum_{i=1}^{N_A} (EAT_i - SAT_i) \quad (1)$$

$$\sum_{i=1}^{N_A} x_{ii} = 1 \quad (2)$$

$$\sum_{i=1}^{N_A} \sum_{j=1}^{N_A} x_{ij} = N_A \quad (3)$$

$$\sum_{j=1; j \neq i}^{N_A} (x_{ij} - x_{ji}) \leq 1 \quad \forall i = 1, \dots, N_A \quad (4)$$

$$\sum_{j=1; j \neq i}^{N_A} (x_{ij} - x_{ji}) \geq -1 \quad \forall i = 1, \dots, N_A \quad (5)$$

$$EAT_i x_{ij} \leq EAT_j \quad \forall i, j = 1, \dots, N_A; j \neq i \quad (6)$$

$$(EAT_i + S_{ij}) x_{ij} \leq EAT_j \quad \forall i, j = 1, \dots, N_A; j \neq i \quad (7)$$

$$x_{ij} \in \{0, 1\} \quad \forall i, j = 1, \dots, N_A \quad (8)$$

$$EAT_i \in [SAT_i, SAT_{imax}] \quad \forall i = 1, \dots, N_A \quad (9)$$

$$EAT_i \text{ integer} \quad \forall i = 1, \dots, N_A \quad (10)$$

(1) is the objective function which represents the total delay of aircrafts. Constraint (2) means that it is one and only one aircraft to land firstly. (3) means all aircraft must land. The flow conservation and the elimination of sub sequences are assured by (4) and (5). (6) means that all aircrafts must land after the first has landed. The constraints (7) are to respect the minimum safety time interval S_{ij} between two successive landings i and j . x_{ij} is a binary variable assured by (8). Finally, (9) respects the time windows of arrival aircrafts of the integer values of the effective arrival times.

2) *The transformation into a linear model:* The constraints (6) and (7) in the initial model (section A.1) are quadratics. We will use a change of variables to make the linearity of our model transforming all quadratic constraints into linear constraints.

Theorem : Using the change of variables:

$$y_{ij} = EAT_i x_{ij}, \quad \forall i, j = 1, \dots, N_A \quad (11)$$

Then the quadratic (6) and (7) can be replaced by the following linear constraints:

$$y_{ij} \leq EAT_j - S_{ij} x_{ij} \quad \forall i, j = 1, \dots, N_A; j \neq i \quad (12)$$

$$y_{ij} \leq SAT_{imax} x_{ij} \quad \forall i, j = 1, \dots, N_A \quad (13)$$

$$y_{ij} \leq EAT_i \quad \forall i, j = 1, \dots, N_A \quad (14)$$

$$y_{ij} \geq SAT_{imax} x_{ij} + EAT_i - SAT_{imax} \quad \forall i, j = 1, \dots, N_A \quad (15)$$

$$y_{ii} \leq EAT_j \quad \forall i, j = 1, \dots, N_A \quad (16)$$

$$y_{ij} \text{ integer} \quad \forall i, j = 1, \dots, N_A \quad (17)$$

Proof:

According (11) we have $EAT_j \geq (EAT_i + S_{ij}) x_{ij}$

$$\Rightarrow EAT_j \geq EAT_i x_{ij} + S_{ij} x_{ij}$$

$$\Rightarrow y_{ij} \leq EAT_j - S_{ij} x_{ij} \text{ (constarint 12)}$$

$$\text{We have } SAT_i \leq EAT_i \leq SAT_{imax} \text{ (18)}$$

$$\Rightarrow y_{ij} \leq SAT_{imax} x_{ij} \text{ (constarint 13)}$$

According to the definition of y_{ij} the $y_{ij} \leq EAT_i$

(constraint 14)

According (18) we have $(SAT_{imax} - EAT_i) \geq 0$, and

on the other hand $x_{ij} \leq 1$

$$\Rightarrow (SAT_{imax} - EAT_i) x_{ij} \leq SAT_{imax} - EAT_i$$

$$\Rightarrow y_{ij} \geq SAT_{imax} x_{ij} + EAT_i - SAT_{imax} \text{ (constarint 15)}$$

While $y_{ij} = EAT_i x_{ij}$ then $y_{ii} = EAT_i x_{ii}$

$$\Rightarrow y_{ii} \leq EAT_j \text{ (constarint 16)}$$

Corollary: The constraints (1), (2), (3), (4), (5), (8), (9), (10), (12), (13), (14), (15), (16) and (17) form the new linear integer model of the ASS problem.

III. ANT COLONY OPTIMIZATION TO THE ASS PROBLEM

Ant Colony Optimization (ACO) is an efficient meta-heuristic proposed by Colormi and Dorigo in 1992 to solve combinatorial optimization problems. The essential trait of ACO algorithms is the combination of a priori information about the structure of a promising solution with a posteriori information about the structure of previously obtained good solutions. Naturally, ants deposited chemical quantity, called pheromone, on the tracks they borrowed. Pheromone are visible by ants and therefore the more a road is used by ants, there are more deposited pheromone and then it becomes more attractive to other ants.

Walking from the nest to the food source and vice versa (which initially is essentially random), ants deposit on the ground in passing a substance fragrant called pheromones. This material allows now to create a chemical trail, on which the ants found there. Indeed, other ants can detect pheromone using sensors on their antennae. Pheromones have a role as a marker of way: when the ants choose their path, they tend to choose the track that carries the highest concentration of pheromones. That their allows to find the way to their nest upon return. On the other hand, the odor can be used by other ants to find food sources found by their peers. This behavior allows you to find the shortest path to food when the tracks pheromones

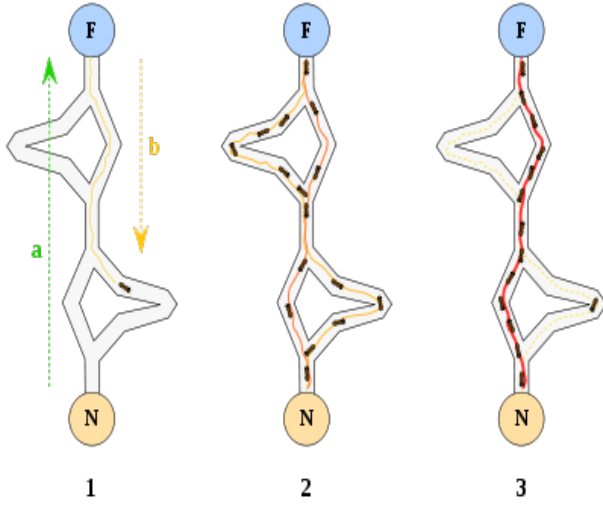


Fig. 2. Principle of ACO [11]

are used by the entire colony. In other words, when multiple paths are marked at the disposal of an ant, the latter may know the shortest way to its destination. Figure 2 give an example for the behavior of colony:

- 1) The first ant finds the food source (F), via any way (a), then returns to the nest (N), leaving behind a trail pheromone (b)
- 2) Ants indiscriminately follow four possible ways, but the strengthening of the runway makes it more attractive as the shortest route.
- 3) Ants take the shortest route, long portions of other ways lose their trail pheromones.

The ant colony algorithm was originally used primarily to produce near optimal solutions to the traveling salesman problem, and, more generally, combinatorial optimization problems. Figure 3 give the conduct of ant colony algorithm for travelling salesman problem. 1) step an ant chooses a path, and draws a trail pheromone, 2) all ant traverses a number of trips, each ant depositing an amount of pheromone proportional to the quality of the course, 3) each edge of the best path is more enhanced than others, 4) evaporation removes the bad solutions. The general structure of ant colony algorithm is given by Algorithm 1.

Algorithm 1 pseudo-code of ACO

```

Initialize pheromone
while not-termination do
    Build solutions components by components.
    Use a heuristic (optional).
    update pheromone trails.
end while

```

In this work, the main principle of ACO algorithm can be summarized as follows. A set of n_f ants (colony of ants) moves through states of the ASS problem. Each ant f

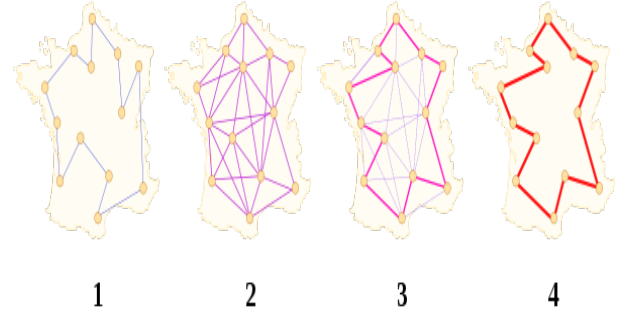


Fig. 3. ACO for Travelling Salesman Problem [11]

corresponds to a solution of ASS problem; f has two vectors as the example in table 1.

order	2	3	4	1	6	5
RAT	100	152	178	203	235	255

TABLE I
AN ANT f WITH 6 AIRCRAFTS

Algorithm 2 ACO for ASS problem

```

Find  $n_f$  initial solutions and calculate  $bestR$ 
Initialize  $\tau_{(p,q)} \leftarrow \frac{1}{bestR}, \forall (p,q) \in A \times A, n_f$  empty ants,
 $iter = 1$ 
while  $iter < iter_{max}$  do
    for all  $f = 1, \dots, n_f$  do
         $NY^f \leftarrow A$ 
        Choose randomly  $p$  by  $f$ ,  $NY^f \leftarrow NY^f \setminus \{p\}$ 
        while  $NY^f$  is not empty do
            Choose a task  $q$  for  $f$  by (20)
             $NY^f \leftarrow NY^f \setminus \{q\}$ 
        end while
        Calculate vector  $EAT$  using formula (19) and  $R^f$ 
        if  $R^f < bestR$  then
             $bestR \leftarrow R^f$ 
        end if
    end for
    Do update pheromones by (21) after finding  $n_e$  best ants
end while
Return best solution achieved with cost  $bestR$ 

```

The first vector represents the landing order of the aircrafts; the order is constructed by f (see Algorithm 1). The second vector EAT represents the real times of landing of the aircrafts. For the first aircraft in order we assign $EAT_2 = SAT_2$ and for each j lands after i we calculate EAT_j using formula (19):

$$EAT_j = \max(SAT_j, EAT_i + S_{ij}) \quad (19)$$

We call R^f the delay of the solution obtained by the ant f . $R^f = \sum_{i \in A} R_i = \sum_{i \in A} EAT_i - SAT_i$ where R_i is the delay of the aircraft which equal to the difference between EAT and SAT of i . Algorithm 1 illustrates our method of ACO on ASS problem. The ants move iteratively by applying a stochastic local decision policy based on two parameters, called trails and attractiveness. Trails represents the amount of pheromone filed between two aircrafts i and j , noted $\tau_{(i,j)}$. Attractiveness represents the visibility measure between two aircrafts i and j , noted $\mu_{(i,j)}$ where $\mu_{(i,j)} = \frac{1}{|SAT_j - EAT_i| + S_{ij}}$.

In the first step, we construct n_f initial solutions; the best cost for the obtained solutions is called $bestR$. Between every two aircrafts i and j , firstly we have the same amount of pheromone equal to $\frac{1}{bestR}$. Note by NY^f the set of aircrafts that are Not Yet in the sequence of ant f . In the second step, at iteration t each ant incrementally constructs a solution to the problem. f choose randomly the first aircraft to serve and it chooses j after using the following probability:

$$j = \begin{cases} \arg_{r \in NY^f} [\tau_{(i,r)}(t)]^\alpha [\mu_{(i,r)}]^\beta, & \text{if } q \geq q_0 \\ J, & \text{Otherwise} \end{cases} \quad (20)$$

With $0 < q, q_0 < 1$ and J is an aircraft calculated by the following standard probability:

$$P_{(i,J)}^f(t) = \begin{cases} \frac{[\tau_{(i,j)}(t)]^\alpha [\mu_{(i,j)}]^\beta}{\sum_{r \in NY^f} [\tau_{(i,r)}(t)]^\alpha [\mu_{(i,r)}]^\beta}, & \text{if } j \in NY^f \\ 0, & \text{Otherwise} \end{cases}$$

After each iteration t of the algorithm, i.e., when all ants have completed a solution, trails are updated. Suppose that we have $u_{(i,j)}^f = 1$ if (i, j) is a sub sequence of f and equal to 0 otherwise, then the rule for global updating pheromone is given as follows:

$$\tau_{(i,j)}(t+1) = \rho \tau_{(i,j)}(t) + \sum_{f=1}^{n_e} \Delta \tau_{(i,j)}^f(t) \quad (21)$$

$$\text{with } \Delta \tau_{(i,j)}^f(t) = u_{(i,j)}^f \frac{1}{R^f}$$

where ρ is the evaporation rate of pheromone ($0 < \rho < 1$), n_e represents the number of elitist ants used. The second term represents the amount of pheromone added to iteration $(t+1)$ between aircrafts existing in the best n_e ants.

IV. NUMERICAL RESULTS

We used a Dell precision M6300 workstation with 2.50 GHz processor speed and 3.00 GB RAM to solve the integer linear program. Each instance has 15 minute per resolution, if in this time we do not have a solution we consider that the instance does not have a feasible solution. The whole instance used for our simulations are proposed by J. E. Beasley on the following URL: <http://people.brunel.ac.uk/~mastijb/jeb/orlib/airlandinfo.html>. both attribute freeze time and appearance time are omitted since the static case is treated. The numerical results show that our approach is effective

Instance	Size	Value	Time(s)
airland1	10	53	1.7
airland2	15	77	2.8
airland3	20	94	3.0
airland9	20	168	3.4
airland10	20	203	3.9
airland4	30	8027	3.9
airland5	44	2713	4.5
airland6	50	211	5.3
airland7	100	9559	7.7
airland8	150	23784	12.6

TABLE II
SIMULATION RESULT

especially in computing time. About the solution (Delay cost) we cannot confirm for now, the effectiveness of our method. After a series of experimental tests, we decided to set our parameters as follows:

$$\alpha = 1 = \beta, q_0 = 0.6, n_f = 20, n_e = 7, iter_{max} = 300.$$

V. CONCLUSION

In this work, we have presented a new formulation for the static aircraft arrival sequencing and scheduling, as a quadratic model. A transformation of it into linear integer program was presented by proposing a change of variables to linearize the nonlinear constraints. In the second part of this paper an ant colony algorithm have been proposed. The numerical results show that our approach is effective. For this reason we are now preparing an exact method to compare the results of this method with that of meta-heuristic.

REFERENCES

- [1] A. T. Ernst, M. Krishnamoorthy, and R. H. Store, Heuristic and exact algorithms for scheduling aircraft landings, Networks vol. 34, pp. 229-241, 1999.
- [2] G. B. Cheikh, J. Boukachour, A. H. Alaoui, F. E. Khokhi, Méthode hybride pour l'ordonnement d'atterrissage d'avions basé sur une modélisation Job Shop, CPI2007, Rabat, MAROC, 2007.
- [3] H. Pinol, and J. E. Beasley, Scatter Search and Bionomic for the Aircraft Landing Problem, European Journal of Operational Research, vol. 171, pp. 439-462, 2006
- [4] I. Farah, A. Kansou, and A. Yassine, A new Integer Linear Program for Air Traffic Flow Management, ECDC2010, 4-5 September, Kish Island-IRAN, 2010.
- [5] J. Abela, D. Abramson, M. Krishnamoorthy, A. De Silva, and G. Mills, Computing optimal schedules for landing aircraft, Proceeding 12th National ASOR Conference, Adelaide, Austaurlia, pp. 71-90,1995
- [6] J-B. Gotteland, Optimisation du trafic au sol sur les grands aports, PhD Thesis, Institut National Polytechnique de Toulouse, 2004.
- [7] J. E. Beasley, J. Sonander, and P. Havelok. Scheduling aircraft landing at London Heathrow using a population heuristic, Journal of the operational Research Society, vol. 52, pp. 483-493, 2001.
- [8] J. E. Beasley, M. Krishnamoorthy, Y. M. Sharaiha and D. Abramson, Dynamically Scheduling aircraft landing the displacement problem, Workingpaper, available from the first author at The management School, Imperial College, London SW7 2AZ, England, 1995.

- [9] J. E. Beasley, M. Krishnamoorthy, Y. M. Sharaiha, and D. Abramson, Scheduling aircraft landings-The static case, *Transportation Science* vol. 34, pp. 180-197, 2000.
- [10] L. Bianco, P. Dellolmo and S. Giodani, Minimizing Total Completion Time Subject to Release Dates and Sequence Dependent Processing Times, *Annals of Operations Research*, vol. 86, pp. 393-415, 1999
- [11] Wikipedia: http://fr.wikipedia.org/wiki/Algorithme_de_colonies_de_fourmis (10.15.2011)